Smart Parking Availability System

Sprint 1 — Progress Report

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🎯 Sprint 1 Objective

- Wire ultrasonic sensor
- Measure distance reliably
- Convert distance → Occupied / Available
- Connect ESP32 to WiFi
- Show live demo on phone

Focus: Hardware + Live IoT validation

🔧 Hardware Integration

Components Used

- XIAO ESP32C6
- HC-SR04 Ultrasonic Sensor
- Logic Level Converter
- Breadboard & Jumper Wires

Key Learning

- Breadboard power rails (5V & GND)
- Safe wiring practices
- Voltage differences (5V vs 3.3V)

✓ Physical system assembled correctly

⚡ Voltage Protection (Critical)

Problem

- HC-SR04 ECHO outputs 5V
- ESP32 GPIO accepts 3.3V only

Solution

- Used logic level converter on ECHO
- TRIG connected directly (3.3V safe)

✓ Prevented permanent hardware damage

📏 Sensor Validation

- Used C++ and `pulseIn()`
- Measured echo duration
- Converted to cm

Formula

Distance = duration × 0.034 / 2

✅ Real-time distance verified
✅ Object detection confirmed

🌐 ESP32 WiFi + Live Dashboard

- Installed ESP32 board package
- Connected to 2.4GHz mobile hotspot
- Hosted HTTP WebServer on port 80
- Generated dynamic HTML page

Live demo:

http://172.20.10.3

✅ Real-time parking status visible

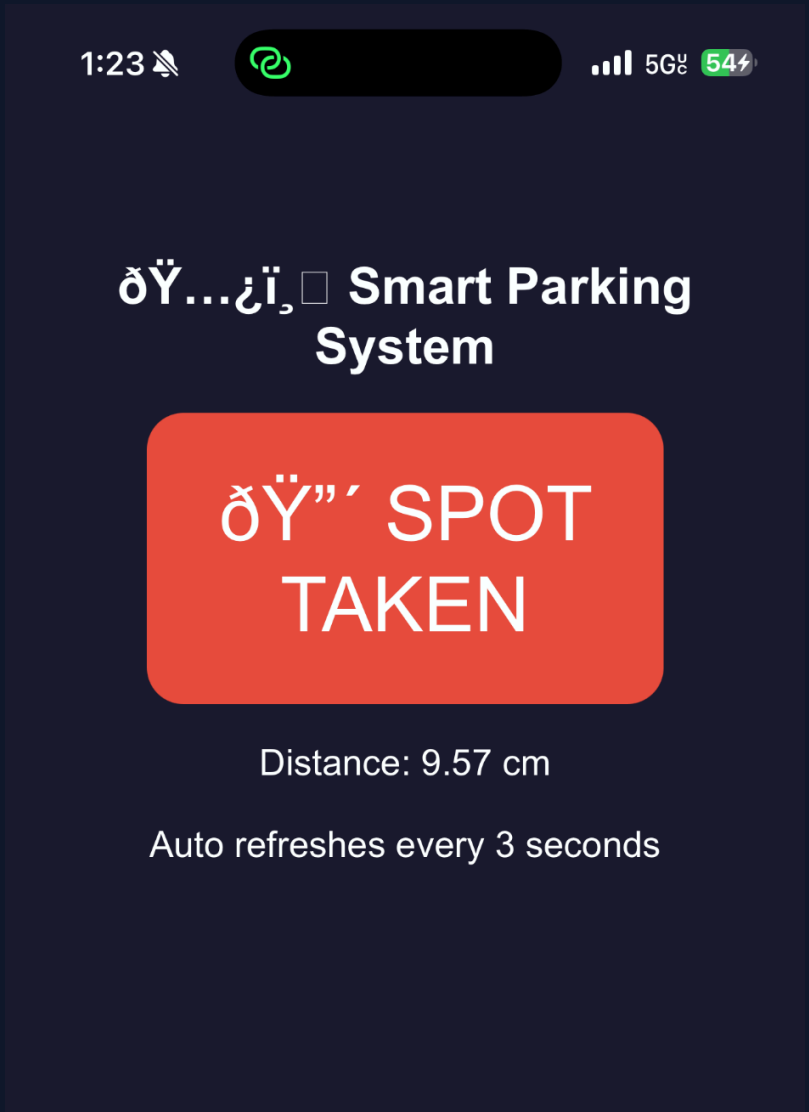
🖥️ Dashboard Output

- 🟢 AVAILABLE
- 🔴 OCCUPIED
- Live distance in cm
- Auto-refresh every 3 seconds

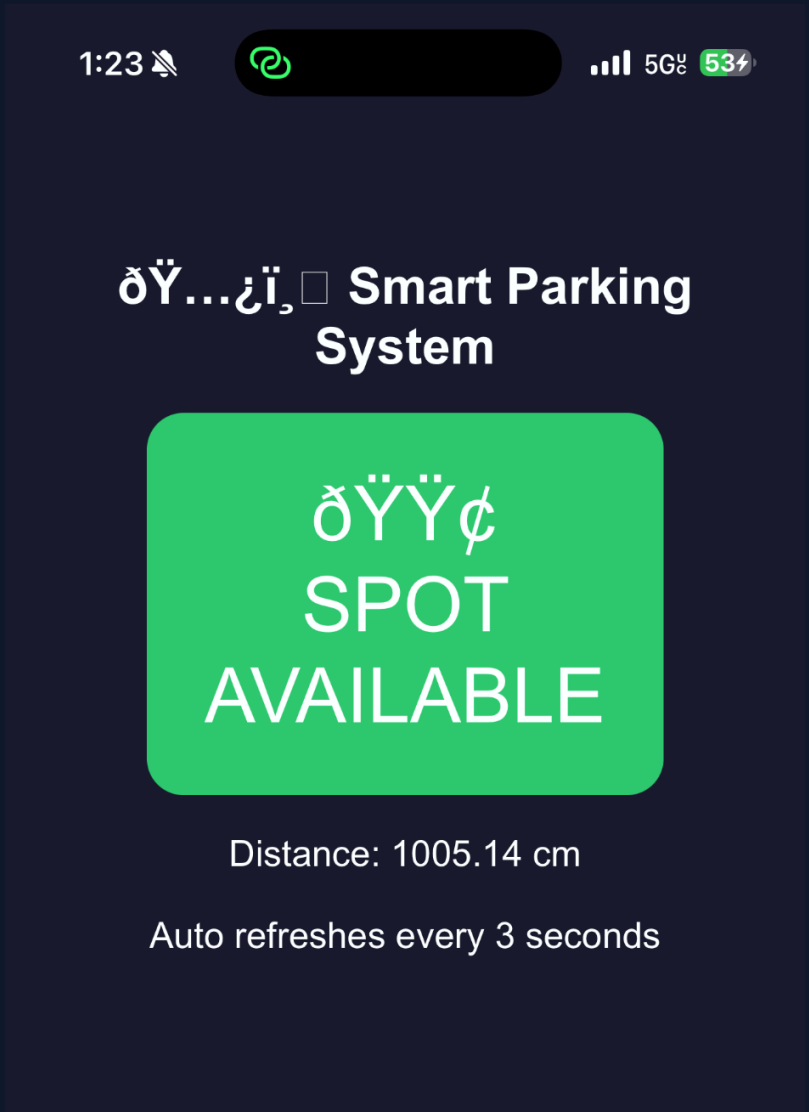
System Flow:

Ultrasonic → ESP32 → WiFi → Phone Browser

Live Demo — Spot Taken



Live Demo — Spot Available



🧠 Two Topics Learned with AI

1 Hardware Systems

- Ultrasonic TRIG/ECHO timing
- Level shifting (5V → 3.3V)
- Breadboard power distribution

2 Embedded Software & IoT Networking

- C++ microcontroller programming
- Threshold detection logic
- WiFi troubleshooting
- Hosting HTTP server on ESP32
- Debugging connection status codes

💪 Major Challenges

- Level shifter wiring confusion
- ESP32 GPIO mapping differences
- WiFi hotspot not broadcasting properly
- Upload and port detection issues
- WiFi password errors (status 6)

📊 Current System Status

Component	Status
Ultrasonic Sensor	✅ Working
Level Converter	✅ Protecting ESP32
ESP32C6	✅ Reading + Serving
WiFi	✅ Stable
Web Dashboard	✅ Live
Detection Logic	✅ Calibrated

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⏭️ Sprint 2 Preview

- Node.js + Express API
- MongoDB database
- React dashboard
- Cloud deployment

Goal: Access from anywhere

Thank You

Questions?